



COMP SCI 354 Syllabus Fall 2025 (LEC 001 & 002)

Institution Name: University of Wisconsin – Madison

Course: Computer Sciences 354 – Machine Organization and Programming

Cross Listed: COMP SCI 354, ECE 354

Credits: 3.00

General Education: Quantitative Reasoning Part B

Breadth: Natural Science

Level: Intermediate

L&S Credit Type: Counts as Liberal Arts and Science (LAS) credit in L&S

Instructors

LEC 001

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LEC 002

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Course Description

An introduction to fundamental structures of computer systems and the C programming language with a focus on the low-level interrelationships and impacts on performance. Topics include the virtual address space and virtual memory, the heap and dynamic memory management, the memory hierarchy and caching, assembly language and the stack, communication and interrupts/signals, assemblers/linkers and compiling.

Prerequisites

COMP SCI/ECE 252 and (COMP SCI 300 or 302) or graduate/professional standing or declared in the Capstone Certificate in Computer Sciences for Professionals as approved through governance, presented in [the Guide](#).

Course Workload Expectations and Credit Hours Fulfillment

This class meets for two 75-minute class periods each week during the semester and carries the expectation that students will work for an *average of 9 hours per week outside of the classroom* on course learning

activities. These activities include reading textbooks, solving programming projects, completing homework quizzes and weekly activities, etc.

Lectures

LEC 001: TR 11:00 AM – 12:15 PM, 1570 Morgridge

LEC 002: TR 2:30 PM – 3:45 PM, 1570 Morgridge

- Attend ALL lectures in person to the best of your ability. Learning via livestream or recording is **not** recommended unless you are sick and can't attend in person.

Canvas Course Website: <https://canvas.wisc.edu/courses/463513>

All course material, programming projects, homework quizzes, weekly activities, grades and the latest news will be posted on the Canvas course page. It is your responsibility to check the page regularly to stay updated with the course.

Instructional Tools

Various tools and platforms will be used as part of the coursework. This includes Canvas, Google Docs, [MobaXterm](#), [PuTTY](#), Quizlet, Zoom, FlipGrid, Microsoft Teams, etc. If you create accounts for any of these tools, be sure to use your *wisc.edu* email address to ensure access to course materials and discussions, as well as to receive credit for graded components of the course.

Course Staff

- **Instructors:** The instructors will hold weekly office hours for student consultations. See office hours schedule on the Canvas course page. Students are also welcome to ask questions in person after lectures (time permitted) or meet with the instructor by appointment.
- **Teaching Assistants (TAs):** 11 TAs are available to assist students with coursework during TA Consulting Hours, starting in Week 1 and continuing through the last day of classes. See complete schedule on the Canvas course page [CS354 Help](#).
- **Peer Mentors (PMs):** 8 PMs are available to assist students with lecture content and coursework during Peer Mentoring Hours, starting in Week 2 and ending the last day of classes. PMs may also plan Bring Your Own Laptop (BYOL) events. See complete schedule on the Canvas course page [CS354 Help](#).

Learning Outcomes

(Only high-level, there are many dozens of learning and coding objectives and skills too)

- Implement and interpret C programs using standard tools and relate C language constructs to both x86 assembly language and fundamental computer system structures.
- Differentiate the memory segments of a process's virtual address space and explain how each is used in C programs.
- Describe and diagram how the heap works internally, analyze the performance of heap allocation strategies, and implement in C a dynamic memory manager.
- Demonstrate how the memory hierarchy functions, differentiate different memory cache implementations, and appraise the effects of memory caching on the performance of C programs.
- Diagram a stack trace of execution for C programs and explain how the compiler implements the stack with x86 assembly language code.

- Formulate C programs that send and receive signals and respond to exceptional circumstances and explain the underlying mechanism that enables asynchronous execution.
- Identify and summarize the steps of C program compilation and describe the processes of linking object code modules to form an executable and loading executables to run them.

Textbooks

- B.W. Kernighan & D.M. Ritchie, [*The C Programming Language*](#), 2nd Edition, Prentice Hall, 1988.
- R.E. Bryant & D.R. O'Hallaron, [*Computer Systems: A Programmer's Perspective*](#), 2nd Edition, Addison Wesley, 2010.

Required Tools

- A terminal application that can be used to remotely connect to [CS Department's Computer Systems Lab \(CSL\)](#) Linux workstations. Students with newer computers (less than two years old) can typically use their built-in **Terminal** application. Students with a terminal application that does not support Linux commands like **ssh** and **scp** can install [MobaXterm](#), [PuTTY](#), or another application that permits secure shell (SSH) remote connections.
- See campus resource [Computers & equipment for students: What do I need?](#)

Weekly Lecture Plan (Tentative)

Week	Dates	Topics	Readings
1	09/03 – 09/07	Introduction and C programming language	K&R: Ch. 1 – 4
2	09/08 – 09/14	Arrays and pointers	K&R: Ch. 5
3	09/15 – 09/21	Multidimensional arrays and structs	K&R: Ch. 5, 6
4	09/22 – 09/28	Input/Output, globals, and virtual memory	K&R: Sec. 7.1, 7.2, 7.4, 7.5 B&O: Sec. 9.1, 9.2
5	09/29 – 10/05	Heap and dynamic memory access MIDTERM EXAM 1	B&O: Sec. 9.9, 9.9.1 – 9.9.6
6	10/06 – 10/12	Heap free list, improvements and caveats	B&O: Sec. 9.9.7 – 9.9.14
7	10/13 – 10/19	Locality and basic cache design	B&O: Sec. 6.2, 6.3, 6.4, 6.4.1, 6.4.2
8	10/20 – 10/26	Caches and more caches	B&O: Sec. 6.4.3 – 6.4.7, 6.6.1
9	10/27 – 11/02	Assembly and machine code	B&O: Ch. 3 Intro Sec. 3.1 – 3.6
10	11/03 – 11/09	Branching and functions in Assembly MIDTERM EXAM 2	B&O: Sec. 3.1 – 3.6
11	11/10 – 11/16	Stack frames, arrays, structs and unions in Assembly	B&O: Ch. 3.7 Intro Sec. 3.7.5, 3.8, 3.9
12	11/17 – 11/23	Flow of execution, exceptions	B&O: Sec. 3.10, 3.12
13	11/24 – 11/30	Process signals Thanksgiving Break	B&O: Sec. 8.1 – 8.4

14	12/01 – 12/07	Multifile compilation, static linking, and ROF	B&O: Sec. 8.5 Intro, 8.5.1 – 8.5.4, 7.1 – 7.8
15	12/08 – 12/10	EOF and Loading	B&O: Sec. 7.9
16	12/17	FINAL EXAM	

COURSEWORK

Graded Components

In-Person Exams	59%	2 midterm exams (15% + 20%) and 1 final exam (24%)
Programming Projects	30%	6 programming projects (some projects have multiple parts)
Homework Quizzes	9%	9 homework quizzes
Weekly Activities	2%	Best 10 out of 12 activities
Total	100%	

Letter Grades

Final letter grades will be determined based on exam performance and the grades earned for all graded components of the course. The table, shown on the right, is *just an estimate and a common starting point* to assign the final letter grades.

The instructor reserves the right to adjust this distribution based on student performance, as it may vary from semester to semester.

A	93% – 100%
AB	88% – 92.99%
B	80% – 87.99%
BC	75% – 79.99%
C	65% – 74.99%
D	55% – 64.99%

In-Person Exams (59%)

In-person proctored exams are the primary method of evaluating student performance in this course. Students must bring their UW PHOTO ID WISC CARD to all exams. All exams are closed book, but references will be provided as necessary. The duration of each exam is **two hours**. Be sure to carefully follow the instructions provided for each exam.

There are two midterm exams and one final exam for this course.

- **Midterm Exam 1 (15%):** Week 5, Wednesday, October 1, 2025, 7:30 PM – 9:30 PM
- **Midterm Exam 2 (20%):** Week 10, Wednesday, November 5, 2025, 7:30 PM – 9:30 PM
- **Final Exam (24%):** Wednesday, December 17, 2025, 12:25 PM – 2:25 PM

Format: The exams may consist of multiple parts.

- Simple choice questions: Cover terms and concepts that mainly require memorization.
- Multiple choice questions: Require code tracing, diagramming, and concepts that require multiple steps and more time to answer.
- Written questions: May include fill-in-the-blank, diagrams, code writing, and other computational problems.

Midterm exams are not cumulative, except as required for understanding new content. Final exam is cumulative with more focus on new material (typically 60-70%) and less focus on prior material (typically 40-30%).

[All exam conflicts must be reported by the end of Week 2 using this link.](#)

Projects (30%)

Programming projects provide an opportunity to develop and enhance your programming, testing and debugging skills, and apply concepts learned in each unit of the course. Concepts required for each project are typically covered before it is assigned so you can begin working on it as soon as it is released. Please note that students who rely on external sources such as internet searches, AI tools, friends, classmates, tutors, and other groups to complete their projects do **not** gain the skills required to solve exam problems or demonstrate solutions to similar, but different, problems on the exams.

- Projects are published on Canvas.
- All projects must be completed **individually**. While you may discuss project-related concepts covered in lectures and textbooks, you are **not allowed** to show or discuss your project code with others. You must develop your own pseudocode or fully report and cite where you got your pseudocode from. Any assistance you receive must be documented and submitted with your project.
- All C programming projects must be done using Linux machines managed by the [CS Department's Computer Systems Lab \(CSL\)](#). You can find your CSLOGIN at the link given below:

<https://apps.cs.wisc.edu/accountapp/>

Login remotely using your Terminal application and the **ssh** command or using an **ssh** client such as [MobaXterm](#). Secure copy files from remote to your local machine using the **scp** command or MobaXterm.

- Submit your completed project work by copying files from your Linux account to your local machine, and then uploading them to Canvas. The last section of each project specifies what to submit and how to check your submission.
- To succeed, start projects as soon as they are released and well before their deadlines. If you encounter challenges, reach out to the course staff for guidance. We can help direct you to appropriate course materials or provide additional resources and support as needed.
- **Document** each session you spend working on your program code within the comments of your program.
- **Students are strongly discouraged from using online searches and AI tools to complete their work.** While these tools may provide answers, the obtained information and results might be incorrect, incomplete, inaccurate and misleading. More importantly, relying on such tools often prevents students from acquiring and building necessary skills that are critical for success in exams.
- **Students must document all sources of help—whether from people or online resources—in their source code.** This practice helps both you and us understand what external assistance you needed and when. Topics requiring outside help often indicate areas that need additional effort on student's part to fully understand the material.
- Students who choose to use internet searches or AI tools must document their use as follows:
 - Include queries in the file header comment of your project files.
 - Save transcripts of any AI “conversations” as a .pdf or .txt file. These files must be provided for review when asked.
 - Failure to document and report such sources will be considered **Academic Misconduct**.
- Questions about your code are best handled during TA or PM consultation hours and **must not** be posted on Piazza or any other course forums.
- All projects are graded on the CS Department's Instructional Linux Computers. Before submitting your projects on Canvas, make sure your work runs correctly as described on CSL computers.
- Make sure to submit early and back up your progress on each assignment to avoid any issues.

- Projects are **not accepted** after Canvas Availability Date. This is because the due dates are already set a week after students are expected to have completed each project. Working on any project past its due date means you are falling behind on the course content and will not be able to work on the current topics and assignments.
- Every project has a few **ontime points**. Submit your completed work by the due date to get your ontime points. Submit your work by the late deadline for other credit, but no ontime points. For one exceptional situation, you can use a one-time **“Oops” point** to submit work shortly after the late deadline. However, please note that you will **lose 1% (20 points) of your total grade** to use this one-time option.

NOTE: Project due dates are managed by Canvas and are clearly displayed on the Coursework page and on the sidebar on the Course home page.

Homework Quizzes (9%)

9 homework quizzes are designed to give you the opportunity to practice answering questions in a manner similar to what you will be expected to do on the exams. Complete each quiz, using **Google Chrome** web browser, as if you are taking an exam. Be sure to trace the code manually and draw and label diagrams. Think through each quiz question without relying on your notes, textbooks, computers, or others. Although you are allowed to use notes and books on quizzes, you will **not** be permitted to do so during exams.

- Homework quizzes, available on Canvas, must be completed **individually**.
- Students are typically given more than one attempt to complete each quiz. The highest score will be recorded as the final score for that quiz.
- All quiz attempts must be completed before the availability end date on Canvas. Students will **not** be able to see the questions otherwise.
- Any quiz that is completed after the deadline and before the availability end date will be penalized 10% for lateness.
- Quiz solutions will be available on Canvas after the quiz availability period has ended.
- While you are encouraged to collaborate with classmates, PMs and TAs to understand course material (textbook topics, lecture concepts, and examples), you are **not permitted** to discuss, share, or exchange algorithms, pseudocode, or code with others. If you have a question, refer to your textbooks or notes for the answer.
- Students who rely on external sources such as internet searches, AI tools, friends, classmates, tutors, and other groups to complete their Canvas quizzes do not gain the skills required to solve exam problems or demonstrate solutions to similar, but different, problems on the exams.

NOTE: Quiz due dates are managed by Canvas and are clearly displayed on the Coursework page and on the sidebar on the Course home page.

Weekly Activities and Oops Point (1% + 1%)

12 weekly activities include pre-lecture quizzes and simple tasks designed to help you stay current with the course material in between larger programming projects and quizzes. Each week, there will be one or two short tasks to complete in order to earn activity points. You need to earn at least 10 of these activity points to earn an **Oops** point. Students are encouraged to help each other with these weekly tasks while adhering to academic integrity guidelines.

Weekly activities may also include the following:

- *FlipGrid Videos*: Short 1-2 minute videos where you answer a question or demonstrate your understanding of a topic or concept.

- *Quizlets*: Give you a chance to practice and reinforce your understanding of terminology used in the course.
- *Worksheets and Practice Quizzes*: Help you practice and complete problems and activities that you can discuss with your classmates and allow you to gauge if you are keeping pace with the course and understanding the smaller components needed to complete the larger graded tasks.

Oops Point: If you complete activities and earn your Oops point, you will receive 1% in addition to the 1% activity points. Once you have earned your Oops point, you may use it if you fail to submit a project on Canvas before its due date has passed. This can only be **used once** (and must be done within two days of the missed late deadline and 96 hours of the original deadline) so it is best to earn it and never have to use it.

GOOD TO KNOW

How to Succeed in this Course?

- Attend ALL lectures in person and take detailed notes to reinforce your understanding.
- Read the corresponding chapters/sections from textbooks. Textbooks often present information in ways that our lectures and outlines may not cover fully due to time constraints.
- Keep up with the weekly readings to make your study sessions less overwhelming and more effective, as later concepts build on earlier ones.
- Go beyond just reading the outlines and textbooks—aim to develop a thorough understanding of the key concepts. You should be able to describe and summarize concepts and topics in your own words to others.
- Identify areas and concepts you find challenging and pay special attention to them.
- Study with a classmate while adhering to academic integrity guidelines or form a small study group to quiz each other. The best way to learn is to teach!
- Do ALL graded and practice tasks yourself. This is the best and most effective way to learn this course. Using online searches and AI tools to complete your work might earn you points, but this will hold you back from gaining critical-thinking and problem-solving skills required to solve exam problems.
- Start early—begin working on projects, quizzes, and exam preparation early to manage your time effectively and avoid last-minute stress.
- Prioritize your well-being with effective time management and self-care to maintain focus and productivity.
- Do not hesitate to seek clarification or assistance from course staff on what you do not understand.
- Above all, *approach the course with curiosity, open mind and a genuine desire to learn. Embrace challenges as opportunities for growth!*

COURSE POLICIES FOLLOW UNIVERSITY POLICIES

All course policies align with university policies, which are available [here](#).

Academic Calendar & Religious Observances

- For details, see [here](#).
- Report any religious observances to the instructor at the start of the semester. Use the **exam conflicts form** (see link on the Canvas course page) to report any religious observances that will keep you from submitting work on a M-F of the semester.
- Establishment of the academic calendar for the University of Wisconsin-Madison falls within the authority of the faculty as set forth in Faculty Policies and Procedures. Construction of the academic

calendar is subject to various rules and laws prescribed by the Board of Regents, the Faculty Senate, State of Wisconsin and the federal government. For additional dates and deadlines for students, see the [Office of the Registrar's](#) pages. **Students are responsible for notifying instructors within the first two weeks of classes about any need for flexibility due to religious observances.**

Academic Integrity Statement

By virtue of enrollment, each student agrees to uphold the high academic standards of the University of Wisconsin-Madison; academic misconduct is behavior that negatively impacts the integrity of the institution. Cheating, fabrication, plagiarism, unauthorized collaboration, and helping others commit these previously listed acts are examples of misconduct which may result in disciplinary action. Examples of disciplinary [sanctions](#) include, but are not limited to, failure on the assignment/course, written reprimand, disciplinary probation, suspension, or expulsion.

CS/ECE 354 Policy

- It is **academic misconduct** to share your work with others in any form, at any time, whether during or after taking this course. This includes posting your solutions on publicly accessible websites like GitHub, Chegg, Course Hero, PasteBin, and similar platforms. Please note that employers are **not** interested in seeing your personal CS/ECE 354 coding work—thousands of students have similar work, and it is well-known that assistance is often available for such assignments.
- It is **academic misconduct** to copy or use any part of a program written by someone else, except for the code provided by us as skeleton code.
- You **must** document all help you receive, including queries and chat transcripts, from people or online tools such as StackOverflow, ChatGPT, and others.
- You do not need to document information obtained from official course materials, including lecture notes, textbooks, or course website.

Accommodation for Students with Disabilities

The University of Wisconsin-Madison supports the right of all enrolled students to a full and equal educational opportunity. The Americans with Disabilities Act (ADA), Wisconsin State Statute (36.12), and UW-Madison policy ([UW-855](#)) require the university to provide reasonable accommodations to students with disabilities to access and participate in its academic programs and educational services. Faculty and students share responsibility in the accommodation process. **Students are expected to inform faculty of their need for instructional accommodations during the beginning of the semester, or as soon as possible after being approved for accommodations.** Faculty will work either directly with the student or in coordination with the McBurney Center to provide reasonable instructional and course-related accommodations. Disability information, including instructional accommodations as part of a student's educational record, is confidential and protected under FERPA. (See: [McBurney Disability Resource Center](#)).

Course Evaluations

UW-Madison students have the opportunity to evaluate the courses they are enrolled in and their learning experiences through course evaluations. Most instructors use a digital course evaluation tool. In most instances, students receive an official email two weeks prior to the end of the semester, notifying them that anonymous course evaluations are available. Student participation is an integral component of course development, and confidential feedback is important. UW-Madison strongly encourages student participation in course evaluations.

Diversity and Inclusion Statement

[Diversity](#) is a source of strength, creativity, and innovation for UW-Madison. We value the contributions of each person and respect the profound ways their identity, culture, background, experience, status, abilities, and opinion enrich the university community. We commit ourselves to the pursuit of excellence in teaching, research, outreach, and diversity as inextricably linked goals. The University of Wisconsin-Madison fulfills its public mission by creating a welcoming and inclusive community for people from every background – people who as students, faculty, and staff serve Wisconsin and the world.

Mental Health and Well-Being Statement

Students often experience stressors that can impact both their academic experience and personal well-being. These may include mental health concerns, substance misuse, sexual or relationship violence, family circumstances, campus climate, financial matters, among others.

Students are encouraged to learn about and utilize UW-Madison's mental health services and/or other resources as needed. Visit uhs.wisc.edu or call University Health Services at (608) 265-5600 to learn more.

Privacy of Student Records and Usage of Recorded Lectures

See [here](#) for comprehensive information about FERPA (Family Educational Rights and Privacy Act).

TLDR: All course material and lecture recordings are protected intellectual property of UW-Madison. Students officially enrolled in this course may use the course materials and recordings solely for their personal use related to their participation in this class. Students may also take notes solely for their personal use. However, **students may not record, copy, or post any of this material, their notes, or their solutions on public forums or in public repositories.**

Students are not authorized to record lectures without permission unless they are considered by the university to be a qualified student with a disability who has an approved accommodation that includes recording. [Regent Policy Document 4-1]

Students may **not** copy or have lecture materials and recordings outside of class, including posting on internet sites or selling to commercial entities, with the exception of sharing copies of personal notes as a notetaker through the McBurney Disability Resource Center. Students are otherwise prohibited from providing or selling their personal notes to anyone else or being paid for taking notes by any person or commercial firm without the instructor's express written permission. **Unauthorized use of these copyrighted lecture materials and recordings constitutes copyright infringement and may be addressed under the university's policies, UWS Chapters 14 and 17, governing student academic and non-academic misconduct.**

Students' Rules, Rights & Responsibilities

[Rights & Responsibilities](#)

Teaching & Learning Data Transparency Statement

The privacy and security of faculty, staff and students' personal information is a top priority for UW-Madison. The university carefully reviews and vets all campus-supported digital tools used for teaching and learning, including those that support [data empowered educational practices](#) and proctoring. View the university's full [teaching and learning data transparency statement](#).

Student Support Services

Do [report any bias or hate incident](#) you witness or experience.

Seek help whenever you need it. The earlier you reach out for advice, support and assistance, the sooner you will be able to get access to resources that can help you. These services are designed to support you and connect you with other resources, as well as with campus and computer science communities that value and honor your talents.

- [University Health Services](#)
- [Undergraduate Academic Advising and Career Services](#)
- [Office of the Registrar](#)
- [Office of Student Financial Aid](#)
- [Student Assistance and Support](#)
- [Food Assistance](#)
- Some [Student Organizations](#) you may wish to connect with:
 - [National Society of Black Engineers – Wisconsin](#)
 - [American Indian Science and Engineering Society \(UW-Madison Chapter\)](#)
 - [Gender and Sexuality Campus Center](#)
 - [WACM \(UW-Madison Chapter of ACM's Women in Computing\)](#)